FINAL PROJECT REPORT

Customer Segmentation using Unsupervised Learning

1. PROBLEM UNDERSTANDING

1 Telecom companies experience high customer churn which reduces revenue and growth.
2 Acquiring new customers is 5–7 times more expensive than retaining existing customers.
3 Generic retention strategies fail because customers have different behaviors and needs.
4 Data■driven behavioral segmentation is required to design targeted retention strategies.

Why Unsupervised Learning?

1 No labeled data available for natural customer groups.
2 Discovers hidden behavioral patterns.
3 Supports proactive retention strategies.
4 More exploratory than supervised learning.

2. APPROACH

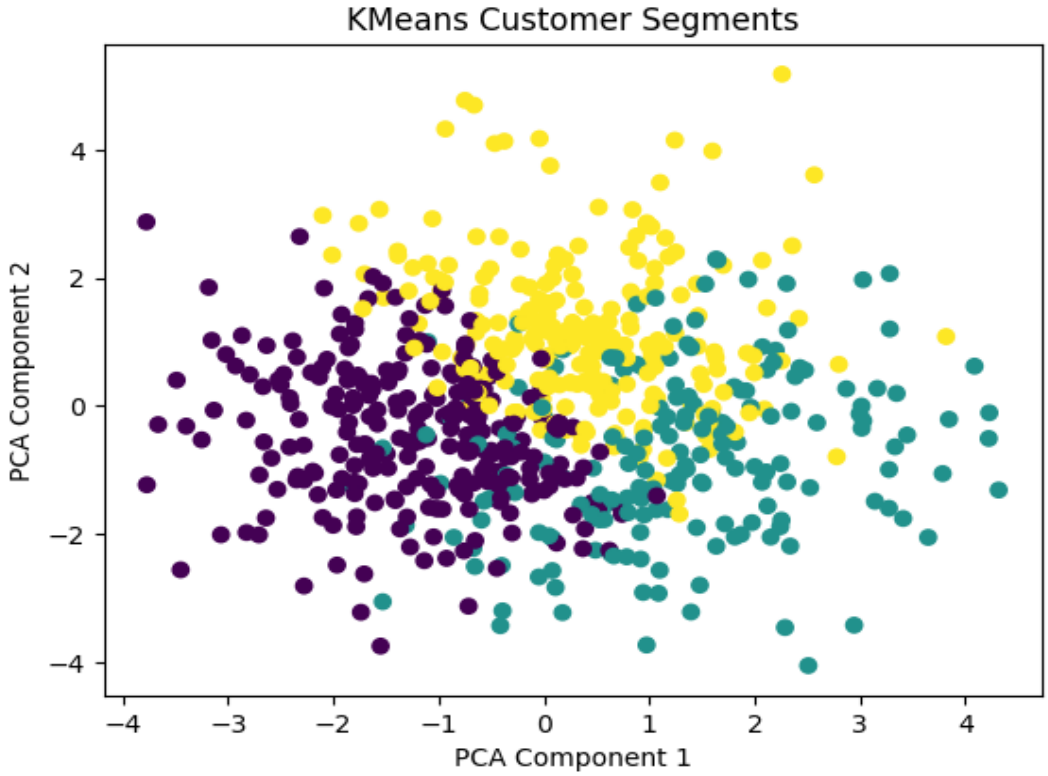
1 Data collection from telecom churn dataset.
2 Data preprocessing and encoding categorical features.
3 Feature scaling using standardization.
4 Dimensionality reduction using PCA.
5 Clustering using KMeans, DBSCAN, and Hierarchical clustering.
6 Cluster evaluation and interpretation.
7 Business insight generation.

3. ALGORITHM COMPARISON

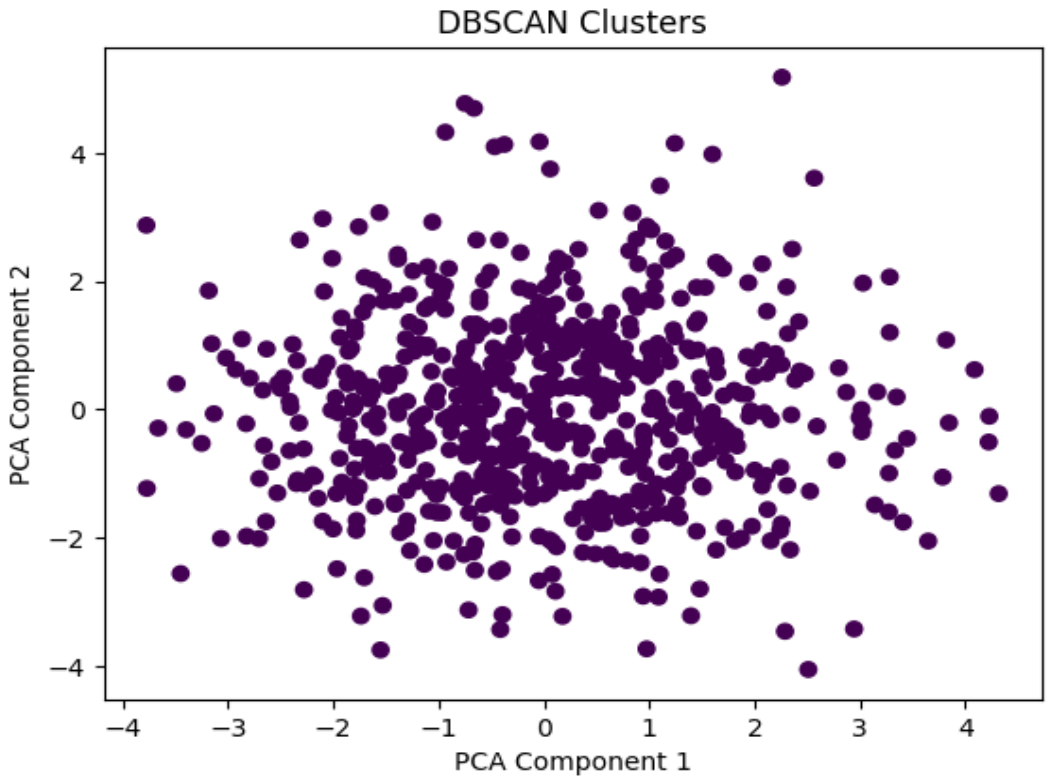
KMeans achieved the best performance with strong cluster separation and stability. Hierarchical clustering provided useful structural visualization but was computationally heavier. DBSCAN identified noise and outliers but produced less consistent cluster boundaries. Silhouette score evaluation confirmed KMeans as the most effective segmentation model.

4. GRAPHS AND VISUALIZATIONS

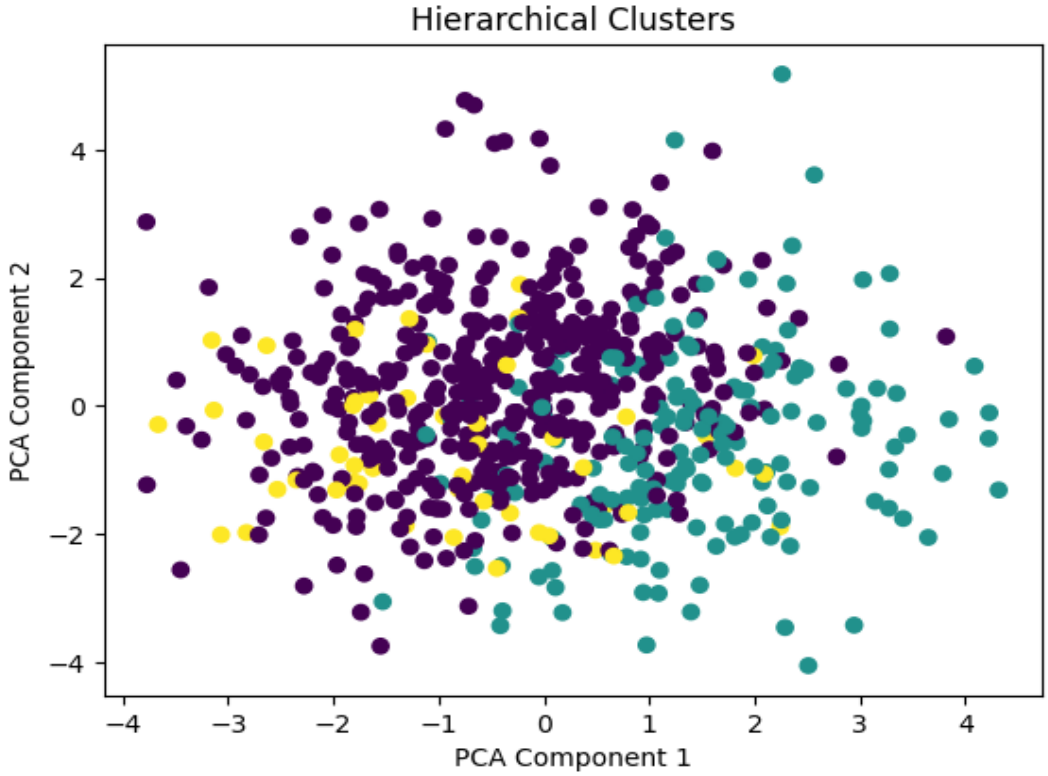
KMeans Cluster Visualization



DBSCAN Cluster Visualization



Hierarchical Cluster Visualization



Cluster Distribution

Cluster	Number of Customers
0	255
1	186
2	226

5. CLUSTER EXPLANATION

1 Cluster 1 — High■Value Loyal Customers: Long tenure, high usage, low complaints. Suitable for loyalty rewards and premium services.
2 Cluster 2 — Price■Sensitive Moderate Users: Average usage and moderate service interaction. Suitable for flexible pricing and targeted offers.
3 Cluster 3 — At■Risk Low Engagement Customers: Low usage, high service complaints, short tenure. Requires proactive retention intervention.

6. BUSINESS RECOMMENDATIONS

1 Reward loyal high■value customers to increase lifetime value.
2 Offer personalized discounts to price■sensitive customers.
3 Provide proactive support to churn■risk customers.
4 Improve service quality to reduce complaints.
5 Use segmentation for targeted marketing campaigns.
6 Continuously retrain models to improve prediction accuracy.

7. CONCLUSION

Customer segmentation using clustering reveals hidden behavioral patterns in telecom users. KMeans provides the most stable and interpretable segmentation. Segment■based strategies significantly improve retention, campaign performance, and profitability.